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RESEARCH ARTICLE

Explainable AI for Brain Tumor Classification Using Cross-Gated Multi-Path Attention Fusion and Gate-Consistency Loss

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ABSTRACT Accurate and early classification of brain tumors using magnetic resonance imaging (MRI) plays a vital role in guiding timely diagnosis and therapeutic intervention. Traditional convolutional neural networks (CNNs) excel at capturing local anatomical patterns but often struggle with contextual reasoning, while Vision Transformers (ViTs) model long-range dependencies but may overlook fine-grained tumor details. To bridge this gap, we propose a hybrid deep learning framework that fuses ResNet and ViT backbones through a Cross-Gated Multi-Path Attention Fusion (CG-MAF) mechanism. CG-MAF performs dynamic, pixel-wise integration of local and global features using cross-attention and learnable gating, thereby adapting to the heterogeneous texture and spatial complexity of brain tumors. To further enhance interpretability and predictive stability, a gate-consistency loss is introduced, which enforces alignment between dual-branch attentional cues. Evaluated on a composite four-class brain MRI dataset, our model achieves state-of-the-art accuracy of 99.81% and an AUC of 0.9999. Ablation studies confirm the critical role of CG-MAF and gate-consistency loss, while Grad-CAM and Integrated Gradients provide multi-scale visual explanations that validate the model's attention on clinically relevant tumor regions. These findings highlight the potential of our approach for robust and interpretable brain tumor classification. All source code, trained models, and experimental results are publicly available at: https://github.com/MuhammadAhmad7171/Advance-brain-tumor to support full reproducibility.

INDEX TERMS Brain tumor classification, MRI, deep learning, convolutional neural networks (CNN), vision transformer (ViT), hybrid models, cross-gated multi-path attention fusion (CG-MAF), gate-consistency loss, medical imaging, interpretability.

I. INTRODUCTION

Brain tumours remain a major cause of cancer-related mortality, and earlier, more reliable diagnosis is key to improved outcomes [1]. Magnetic resonance imaging (MRI) provides detailed visual evidence of pathological change, yet manual reading is slow and intrinsically subjective [2]. Deep learning (DL) therefore offers an attractive route to automated, high-throughput assessment. Convolutional neural networks (CNNs) lead this effort by learning hierarchical edges, textures and other fine-scale traits that mark tumour borders [3]. Their weakness lies in modelling long-range context across the brain volume. Vision transformers (ViTs) address that gap by applying self-attention to every patch

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and thus capturing global structure; however, they often miss subtle local cues and are data-hungry two persistent hurdles in medical imaging [4].

Hybrid approaches that mix CNNs and ViTs have begun to appear, aiming to balance locality and globality. Most, however, rely on simple concatenation schemes that neither weight nor coordinate the two information streams [3], [5]. To advance beyond this stage, we introduce a ResNet–ViT architecture equipped with a Cross-Gated Multi-Path Attention Fusion (CG-MAF) module [6]. CG-MAF uses learnable gates to pass only the most informative signals from each branch, letting the network decide, pixel by pixel, whether fine detail or holistic context is more useful. In parallel, a gate-consistency loss forces both branches to attend to the same anatomical regions, increasing robustness and offering transparent attention maps for clinical audit. Unlike prior CNN–ViT approaches [5], [24] which fuse features without enforcing inter-branch coherence, our framework uniquely integrates a cross-branch consistency constraint. To the best of our knowledge, this is the first brain-tumor MRI classifier to align dual attention maps via a gated consistency loss, marking a novel contribution in this domain.

We test the model on a four-class MRI set (glioma, meningioma, pituitary tumour and normal tissue) and report 99.81 % accuracy with an AUC of 0.9999, surpassing leading CNN-only or ViT-only baselines. Ablation experiments confirm that removing CG-MAF or the consistency term lowers performance and erodes interpretability, underscoring their complementary benefits.

In summary, our contributions are threefold:

CONTRIBUTIONS

- **Dynamic CG-MAF fusion:** Adaptively integrates local (ResNet) and global (ViT) features at pixel-level for precise tumor focus.
- **Gate-Consistency Loss (GCL):** Aligns attentional cues between branches, enhancing interpretability and decision coherence.
- **State-of-the-art performance:** Achieves 99.81% accuracy and 0.9999 AUC, with multi-scale visual explanations validating clinical relevance.
- **Lightweight, deployable architecture:** Eliminates ensemble overhead, ensuring robust and transparent clinical use.

II. RELATED WORK

Recent progress in brain-tumour MRI analysis has centred on deep-learning models that either excel at local texture (CNNs) or capture global context (ViTs). Our work combines both strengths through a Cross-Gated Multi-Path Attention Fusion (CG-MAF) module and a gate-consistency loss, aiming to close the remaining gap between fine-detail discrimination and whole-brain reasoning. To set the stage, we organise earlier studies into four themes and highlight the specific limitations that motivate our hybrid design.

A. DEEP HYBRID MODELS

The integration of convolutional neural networks (CNNs) with Vision Transformers (ViTs) has emerged as a key strategy to address the complementary strengths of local texture extraction and global context modeling in brain tumor MRI classification. Early CNN-based architectures have demonstrated proficiency in capturing fine-grained anatomical details. For instance, *Zhang* et al. achieved 96.7% accuracy using a residual-dense CNN [7], while *Yang* et al. enhanced BraTS 2020 outcomes to 98.3% by incorporating channel attention mechanisms [8]. Lightweight CNN variants, such as the SE-block CNN by Huang and Lee [9] and shallow-wide CNNs proposed by *Lee* et al. [10], further optimized inference efficiency but encountered limitations in capturing global tumor morphology and struggled with rare class generalization. Ensemble CNN strategies like the triple-CNN system from *Patel* et al. pushed accuracy beyond 99% [11], though at a substantial computational cost. Hybrid learning pipelines integrating data augmentation, such as the PGGAN-augmented VGG19 model by *Gab Allah* et al. [12], and transfer learning with EfficientNetB2 and Grad-CAM visualization by *BabuVimala* et al. [13], as well as ensemble frameworks by Khan et al. [53], Anand et al. [54], and recent deep learning pipelines [55], further extended accuracy boundaries. Despite these advancements, static feature concatenation and rigid fusion strategies in prior works constrained adaptive feature relevance, a gap addressed by gated hybrid models. *Khaniki* et al. introduced selective cross-attention in CNN–ViT hybrids, achieving 99.24% accuracy [5], and *Asiri* et al. explored ViT fine-tuning, reporting up to 98.24% accuracy [15]. These studies reinforce the transition towards dynamic fusion frameworks that unify local and global representations, setting the stage for architectures like our proposed Cross-Gated Multi-Path Attention Fusion (CG-MAF) mechanism.

B. FEATURE FUSION STRATEGIES

The design of effective feature fusion mechanisms is pivotal in bridging the granularity gap between local convolutional descriptors and transformer-derived global dependencies. Existing literature has explored various static and semi-dynamic fusion approaches. *Gao* et al. introduced PFormer, integrating content-driven P-attention modules within a CNN–Transformer hybrid, achieving a Dice Similarity Coefficient (DSC) of 86.87% [15]. *Liu* et al. proposed ConvBiFuseNet, employing bi-level routing attention to synchronize feature pathways from ConvBraNet and Biformer modules, resulting in 98.36% accuracy on CE-MRI datasets [16]. Traditional fusion methods combining CNN-extracted features with classical classifiers, such as the SVM-based pipeline by *Yefan Liu* et al., achieved high validation accuracy (99.94%) but lacked adaptive modulation during inference [17]. *Khaniki* et al. advanced selective cross-attention fusion within ViT backbones to refine feature

calibration, reaching 99.24% accuracy [5]. Unlike prior rigid fusion schemes, our CG-MAF enables dynamic, gated integration of local and global features for adaptive and robust tumor classification.

C. EXPLAINABLE AI

The interpretability of deep learning models is a critical enabler for clinical adoption, necessitating the integration of explainable AI (XAI) frameworks within brain tumor MRI classification pipelines. Early explainability efforts, such as BMRI-Net by *Mondal and Shrivastava*, utilized customized activation functions (PFpM) to enhance feature saliency [18]. *Saeed* et al. developed Neuro-XAI, combining DeepLabV3+ with Bayesian Optimization and Grad-CAM to visualize diagnostic regions [19]. Multi-explanation strategies, exemplified by *Akgündoğdu and Çelikbaş*, fused LIME, Grad-CAM, and SHAP for enriched interpretability in dual-stage CNN models [20]. Attention-driven visualization techniques, such as Grad-CAM employed by *Tonmoy* et al. in their X-Brain architecture, achieved interpretable predictions with 99.30% accuracy [21]. Lightweight XAI-enabled CNN designs by *Iftikhar* et al. [22] and Grad-CAM augmented EfficientNetB2 pipelines by *BabuVimala* et al. [13] further emphasized the necessity of transparent feature attribution. While these approaches provided post-hoc interpretability, they often lacked inherent alignment within the learning process. In contrast, our methodology integrates a Gate-Consistency Loss (GCL) that enforces spatial attention coherence across the CNN and ViT pathways, inherently producing reliable saliency maps validated through Grad-CAM and Integrated Gradients.

D. GATED MECHANISMS AND CONSISTENCY REGULARIZATION

Adaptive gating and consistency mechanisms have emerged as transformative elements in enhancing model robustness and interpretability in brain tumor analysis. *Huang* et al. presented UDNet, unifying CNN and Transformer pathways via Multi-Stage Fusion (MSF) and Inter-Active (IA) modules, attaining 95.88% accuracy across diverse datasets [23]. *Zhou* et al. introduced a Gated Dual-Branch Hybrid Attention framework, blending Multi-Scale Cross-Attention and Self-Attention mechanisms, achieving notable Dice scores of 91.97% [24]. Feature Calibration and Selective Cross-Attention techniques by *Khaniki* et al. within ViT backbones further emphasized the relevance of adaptive gating in model design [5]. However, these models primarily focused on attention refinement without enforcing inter-branch consistency during feature fusion. Addressing this gap, our proposed Gate-Consistency Loss (GCL) synergizes with CG-MAF to ensure alignment of attentional cues between ResNet and ViT branches, leading to superior interpretability and classification reliability. This joint formulation enables our architecture to dynamically regulate feature contributions while maintaining decision consistency, which is crucial for deployment in clinical diagnostic workflows.

III. MATERIAL

A. DATASET DESCRIPTION

In this study, we utilize a publicly available brain MRI dataset sourced from Kaggle [25] for tumor classification tasks. The original dataset is compiled from three widely cited repositories: Figshare Brain Tumor Dataset, Kaggle Brain Tumor Classification MRI, and Kaggle Brain Tumor Detection (Br35H).

The dataset comprises 7,153 MRI slices distributed across four diagnostic categories: glioma, meningioma, pituitary tumor, and no-tumor cases, as detailed in Table 1. An 80:20 split is adopted, with stratified sampling to ensure balanced test sets across all tumor types.

For all experiments, we carefully ensured that the data partitions remained strictly non-overlapping at the subject level. Each cross-validation fold was divided into three distinct subsets: training, validation, and testing. The independent test subset in every fold contained only unseen subjects and was reserved exclusively for the final evaluation. From the remaining pool, a small fraction was separated as validation data to guide model selection and early stopping, while the rest was used for learning. This protocol guarantees that no image slice from a given subject could appear simultaneously in both training/validation and testing, thereby eliminating any risk of data leakage.

Representative examples of axial, sagittal, and coronal MRI slices for each tumor class are illustrated in Figure 1, where tumor regions are highlighted with red bounding boxes.

This composite dataset forms the foundation for evaluating the effectiveness of our Cross-Gated Multi-Path Attention Fusion (CG-MAF) and Gate-Consistency Loss (GCL) mechanisms.

TABLE 1. Per-class statistics for the composite brain-MRI dataset.

Class ID	Tumour type / dir.	# Slices
0	Glioma	1 621
1	Meningioma	1 775
2	Pituitary tumour	1 757
3	No-tumour (healthy)	2 000
Total		7 153

B. PRE-PROCESSING AND DATA AUGMENTATION

All MRI slices were first resized to a standard resolution of 224 × 224 using bicubic interpolation to align with the ImageNet pretraining requirements of ResNet and ViT backbones. For training samples, a rich augmentation pipeline was employed to improve generalization, including random rotations (±20°), horizontal flips, color jittering (brightness, contrast, saturation, hue), perspective distortions, and affine transformations with translation and scale jitter. This was followed by normalization using ImageNet mean and

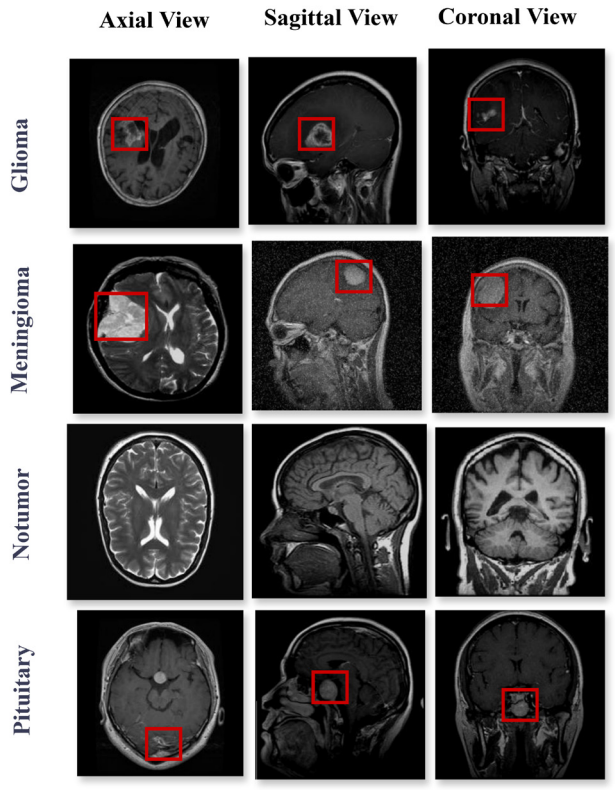


FIGURE 1. Representative axial, sagittal and coronal slices for each class. Red boxes mark tumour cores.

standard deviation statistics [$\mu = (0.485, 0.456, 0.406), \sigma = (0.229, 0.224, 0.225)$]. For validation and test sets, only resizing and normalization were applied to maintain consistency during evaluation. No skull stripping or histogram matching was applied, allowing the model to learn feature invariance under realistic scanner variance.

IV. METHODS

In this section, we describe each component of our CG-MAF architecture: **(1)** the ResNet-50 branch for local feature extraction, **(2)** the ViT-Base branch for global context modeling, **(3)** the Cross-Gated Multi-Path Attention Fusion (CG-MAF) block that dynamically integrates both streams, and **(4)** the gate-consistency loss that aligns branch predictions. We conclude with the training pipeline and key hyperparameters.

A. RESNET-50 BRANCH: LOCAL FEATURE EXPERT

ResNet-50 [26] is a 50-layer deep convolutional network that employs *residual connections* to mitigate the vanishing gradient problem, allowing stable training at large depths. This architecture is widely used for extracting local features from images, particularly for edge and texture-based feature extraction.

Each bottleneck block in ResNet-50 applies three convolutions: a 1×1 reduction layer, a 3×3 spatial convolution, and a 1×1 expansion layer. The block output is computed as:

$$\mathbf{h}^{(\ell+1)} = \mathbf{h}^{(\ell)} + W_3^{(\ell)}\,\sigma\big(\mathrm{BN}(\,W_2^{(\ell)}\,\sigma(\mathrm{BN}(W_1^{(\ell)}\mathbf{h}^{(\ell)}))\big)\big),$$

where $W_k^{(\ell)}$ are weight tensors, BN is batch normalization, and σ is the ReLU activation function. The residual shortcut path $\mathbf{h}^{(\ell)}$ allows gradients to propagate more easily, which is essential for training deep networks.

1) USAGE IN OUR MODEL

After extracting $\mathbf{r} \in \mathbb{R}^{2048}$, we project the features to a 512-dimensional embedding:

$$\mathbf{r}' = W_r\,\mathbf{r} + \mathbf{b}_r, \quad W_r \in \mathbb{R}^{512\times2048}.$$

This lower-dimensional representation $\mathbf{r}'$ is then passed to the CG-MAF block alongside the ViT branch for further processing.

B. VIT-BASE BRANCH: GLOBAL CONTEXT EXPERT

The Vision Transformer (ViT-Base) [27] treats an image as a sequence of patches, processing them with self-attention mechanisms to capture long-range dependencies that are difficult for traditional CNNs to model.

1) *Patch embedding.* Each input image $I \in \mathbb{R}^{3\times224\times224}$ is split into 14×14 non-overlapping patches $P_i \in \mathbb{R}^{3\times16\times16}$, flattened, and linearly projected:

$$\mathbf{e}_i = W_E\,\mathrm{vec}(P_i) + \mathbf{b}_E, \quad \mathbf{e}_i \in \mathbb{R}^{768}.$$

A learnable class token $\mathbf{t}_0$ is added at the beginning of the sequence, along with positional encodings $\mathbf{P}$, forming the input $\mathbf{H}^{(0)} = [\mathbf{t}_0; \mathbf{e}_1; \ldots; \mathbf{e}_{196}] + \mathbf{P}$.

2) *Multi-head self-attention (MHSA).* For each of $L = 12$ transformer layers, the input sequence is passed through multi-head attention:

$$\mathbf{O}_h = \mathrm{softmax}\Big(\frac{\mathbf{Q}_h\mathbf{K}_h^\top}{\sqrt{64}}\Big)\mathbf{V}_h,$$

where $\mathbf{Q}_h, \mathbf{K}_h$, and $\mathbf{V}_h$ are the queries, keys, and values, respectively, and $d_h = 64$. The output of each head is concatenated and passed through a feed-forward network to obtain the final class token $\mathbf{v} = \mathbf{H}_0^{(12)} \in \mathbb{R}^{768}$, representing the global context of the image.

3) *Projection for fusion.* The final output $\mathbf{v}$ is projected to a 512-dimensional space:

$$\mathbf{v}' = W_v\,\mathbf{v} + \mathbf{b}_v, \quad W_v \in \mathbb{R}^{512\times768}.$$

1) WHY IT HELPS

The ViT-Base model excels in capturing long-range relationships between regions of an image. This is particularly useful in medical imaging, where distant anatomical features (e.g., tumours on opposite sides of the brain) need to be considered together.

C. CROSS-GATED MULTI-PATH ATTENTION FUSION (CG-MAF)

CG-MAF is a fusion mechanism that enables both ResNet and ViT branches to vote on the relevance of each other's features and dynamically weights the contribution of each branch for the final feature representation. Unlike prior CNN–ViT fusion schemes that statically combine features without enforcing coordination between streams, CG-MAF performs fine-grained, adaptive integration. It uses learnable gates to transmit only the most informative signals from each pathway, allowing the network to decide, pixel by pixel, whether local detail or global context is more useful for a given region. This dynamic modulation enhances the network's flexibility and robustness when processing heterogeneous tumor textures.

- *Cross-attention probe:* The two embeddings $\mathbf{r}'$ (ResNet) and $\mathbf{v}'$ (ViT) are combined using attention mechanisms:

$$\mathbf{c}_r = \text{softmax}(\mathbf{r}'\mathbf{v}'^{\top})\,\mathbf{v}', \quad \mathbf{c}_v = \text{softmax}(\mathbf{v}'\mathbf{r}'^{\top})\,\mathbf{r}'.$$

  These operations enable each feature set to probe the other for relevance.
- *Gate computation:* Based on the cross-attention outputs, we compute the relevance gates for each stream:

$$g_{r\to v} = \sigma(W_{rv}\mathbf{c}_r + b_{rv}), \quad g_{v\to r} = \sigma(W_{vr}\mathbf{c}_v + b_{vr}),$$

  where σ is the sigmoid function and the gates $g_{r\to v}$ and $g_{v\to r}$ control how much each stream trusts the other.
- *Fusion:* The final fused feature $\mathbf{f}$ is computed as:

$$\mathbf{f} = g_{r\to v} \odot \mathbf{v}' + g_{v\to r} \odot \mathbf{r}', \tag{1}$$

  where $\odot$ denotes element-wise multiplication. This fusion step dynamically combines both streams based on their computed relevance, allowing the network to adapt to the nature of the input image.

By enabling selective, gated integration of ResNet and ViT features, CG-MAF ensures that each modality contributes proportionally to its confidence and relevance. This design promotes coherent feature fusion and prepares the foundation for consistent attention alignment via the gate-consistency loss, ultimately leading to more interpretable and reliable tumor classification.

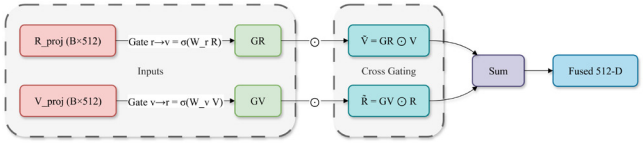


FIGURE 2. CG-MAF Block. Cross-attention is computed between the ResNet and ViT feature maps. The relevance gates decide the contribution of each stream, and the features are fused accordingly.

D. GATE-CONSISTENCY LOSS

The gate-consistency loss is designed to encourage alignment between the ResNet and ViT predictions while weighting the importance of each branch based on its gate confidence. Concretely, we attach an auxiliary classification head to each branch (ResNet and ViT), producing branch-specific logits $\mathbf{z}_r$ and $\mathbf{z}_v$, which are converted into probability distributions p_r and p_v via softmax. In parallel, the fused representation obtained after Cross-Gated Multi-Path Attention Fusion (CG-MAF) is passed through a main classification head to yield the final prediction logits $\mathbf{z}$ and probabilities p. The gate-consistency loss then aligns these outputs by penalizing divergences among them.

Given the logits from the main and auxiliary heads $(\mathbf{z}, \mathbf{z}_r, \mathbf{z}_v)$ and the corresponding probabilities (p, p_r, p_v), the loss is computed as:

$$\mathcal{L} = -\mathbf{y}^{\top}\log p + \frac{\bar{g}_{r\to v}}{2}\,\text{KL}(p_r \parallel p) + \frac{\bar{g}_{v\to r}}{2}\,\text{KL}(p_v \parallel p), \tag{2}$$

where $\bar{g}_{r\to v}, \bar{g}_{v\to r}$ are the mean gate values for the ResNet and ViT streams, respectively, and KL denotes the Kullback-Leibler divergence.

Intuitively, this regularizer compels both the CNN and ViT branches to produce congruent predictions, especially in scenarios where a branch's gate value is highindicating greater confidence in its features. By weighting the divergence terms with the gate magnitudes, the gate-consistency loss ensures that highly confident branch outputs remain consistent with the final fused prediction. This mechanism promotes cross-branch coherence, encouraging the ResNet and ViT pathways to focus on the same anatomical regions, which in turn enhances the stability, robustness, and interpretability of the model's predictions.

E. TRAINING PIPELINE

Figure 3 presents a visual overview of our complete training and evaluation framework, showing each step from data ingestion through to classification and metric reporting. The formal training steps are detailed in Algorithm 1, which outlines the core loop including augmentation, feature extraction, gated fusion, and early stopping.

1) OPTIMISER AND SCHEDULE

We use the AdamW optimiser with an initial learning rate of 1×10^{-4} and a weight decay of 5×10^{-4}, combined with a cosine learning rate scheduler that restarts every 10 epochs.

2) MINI-BATCHING AND REGULARISATION

A physical batch size of 256 is used, with gradients accumulated for $K = 4$ steps to simulate an effective batch size of 1,024. Early stopping is applied if validation accuracy does not improve for 5 epochs, and stochastic weight averaging (SWA) is used starting from epoch 25. Table 2 lists the full hyper-parameter configuration.

3) EXPLAINABILITY

After training, Grad-CAM maps are generated for five random slices per class to ensure that the network attends to the appropriate regions and does not focus on spurious image areas.

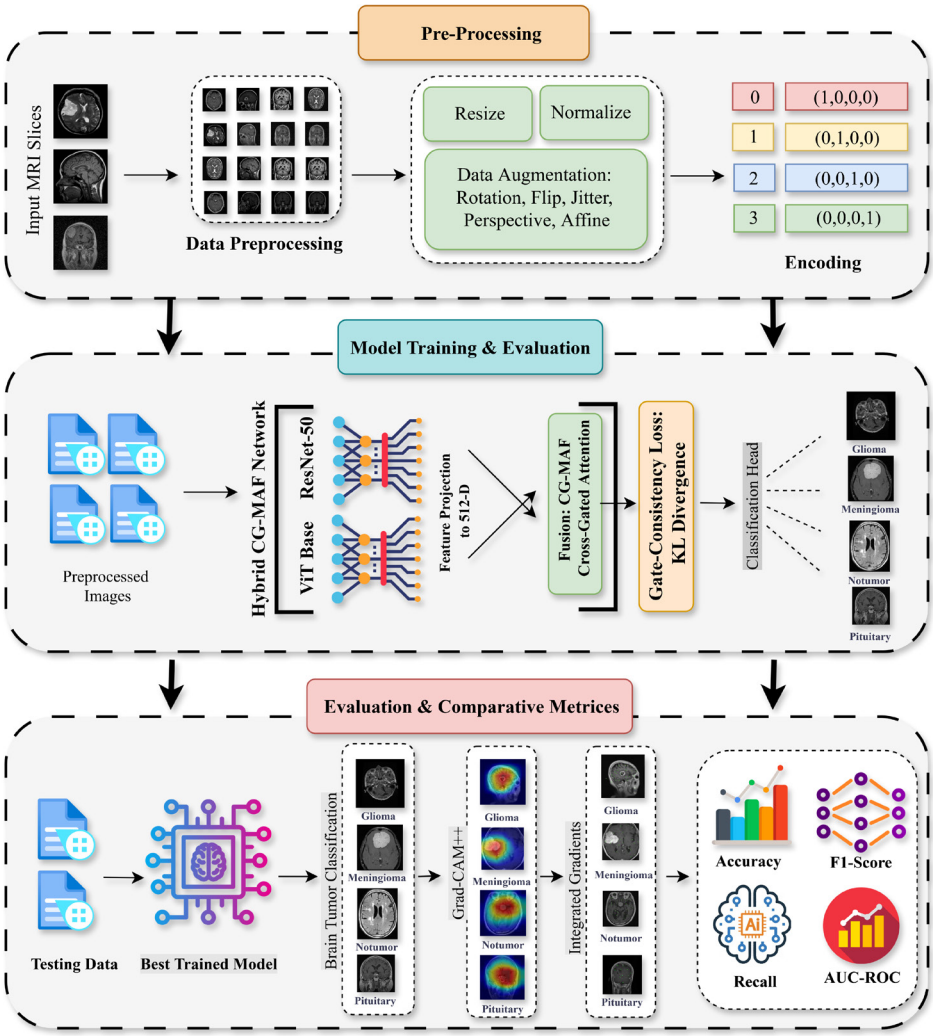


FIGURE 3. End-to-end pipeline of our proposed framework. Top: MRI slices undergo resizing, normalization, and augmentation (rotation, flip, jitter, perspective, affine) before one-hot encoding. Middle: The preprocessed data is passed through a dual-branch hybrid network (ResNet-50 and ViT-Base), fused using CG-MAF with dynamic attention gating and supervised via a gate-consistency loss. Bottom: The trained model is evaluated using standard classification metrics including Accuracy, F1-Score, Recall, and AUC-ROC.

4) IMPLEMENTATION AND REPRODUCIBILITY

All experiments were implemented in PyTorch and executed on a Linux-based server equipped with 7 NVIDIA TITAN RTX GPUs, each with 12GB of VRAM. To ensure full reproducibility, we have made the complete source code, pre-trained model checkpoints, and dataset splits publicly available on GitHub (link provided in the Abstract). Detailed instructions are provided for replicating the entire training pipeline, including environment setup and inference procedures, adhering to best practices for transparent and repeatable research.

F. EVALUATION METRICS

At the end of each fold, we compute a suite of standard classification metrics to fully characterise model performance. Using the test-set predictions and ground-truth labels, we obtain:

- **Overall accuracy,** defined as the fraction of correct predictions, using `(preds == labels).mean()` in NumPy [28].

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{3}$$

- **Precision, recall,** and **F_1-score**, extracted via `classification_report` in scikit-learn [29], which summarises per-class and macro-averaged values.
- **Precision**, the correctness of positive predictions

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{4}$$

- **Recall** (sensitivity), the coverage of actual positives:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{5}$$

Algorithm 1 Proposed CG-MAF Training and Testing Procedure

Input: Training data $\mathcal{D}_{\text{train}} = \{(X_i, y_i)\}_{i=1}^{N}$; Images $X_i \in \mathbb{R}^{3\times224\times224}$; Backbones: ResNet-50 (local), ViT-Base (global); CG-MAF fusion block f_{fuse} (Section IV-C);
Classifier f_{cls}, Gate-Consistency Loss $\mathcal{L}$ (2); Optimizer: AdamW with learning rate 1×10^{-4}, weight decay 5×10^{-4};
Epochs $E = 30$, patience $p = 5$, batch size $B = 256$.
Output: Trained model weights, best validation performance, and test metrics (Accuracy, AUC, F1)
Initialize: Best metric $\mathcal{M}_{\text{best}} \leftarrow -\infty$; counter $c \leftarrow 0$
for $e \leftarrow 1$ **to** E **do**
  Shuffle training data $\mathcal{D}_{\text{train}}$
  foreach *mini-batch* (X, y) *in* $\mathcal{D}_{train}$ **do**
    Apply augmentations to X (rotation, flip, jitter, affine)
    Extract local features: $\mathbf{r} \leftarrow \text{ResNet}(X)$ // Eq. from Sec. IV-A
    Extract global features: $\mathbf{v} \leftarrow \text{ViT}(X)$ // Eq. from Sec. IV-B
    Project features: $\mathbf{r}', \mathbf{v}' \in \mathbb{R}^{512}$
    Fuse via CG-MAF: $\mathbf{f} \leftarrow f_{\text{fuse}}(\mathbf{r}', \mathbf{v}')$ // Eq. (1)
    Compute logits: $(\mathbf{z}, \mathbf{z}_r, \mathbf{z}_v) \leftarrow f_{\text{cls}}(\mathbf{f}, \mathbf{r}', \mathbf{v}')$
    Compute softmax outputs: (p, p_r, p_v) from $(\mathbf{z}, \mathbf{z}_r, \mathbf{z}_v)$
    Compute gate-consistency loss: $\mathcal{L} \leftarrow$ Eq. (2) // Main loss function
    Backpropagate gradients
    if *step % 4 == 0* **then**
      Update parameters via AdamW; zero gradients
    end
  end
  Evaluate model on validation set $\mathcal{D}_{\text{val}}$ ' metric $\mathcal{M}_{\text{val}}$
  if $\mathcal{M}_{val} > \mathcal{M}_{best}$ **then**
    Save model weights; $\mathcal{M}_{\text{best}} \leftarrow \mathcal{M}_{\text{val}}$; $c \leftarrow 0$
  else
    $c \leftarrow c + 1$
  end
  if $c \geq p$ **then**
    Break // Early stopping
  end
end
Final Evaluation: Load best model and compute test metrics: Accuracy, Precision, Recall, F1, AUC, Confusion Matrix

TABLE 2. Key Hyper-parameters.

Parameter	Value
Initial LR η_0	1×10^{-4}
Weight decay λ_w	5×10^{-4}
Cosine restart period T_0	10 epochs
Batch (physical / effective)	256 / 1 024
Epoch budget E	30
Early-stop patience p	5 epochs
Gate-consistency weight	0.5
SWA start epoch	25
GPUs	7 × TITAN RTX 12 GB

- **F_1-score**, the harmonic mean of precision and recall:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{6}$$

- **Confusion matrix**, built with `confusion_matrix` [29] to visualise class-wise true/false positives and negatives.
- **ROC AUC**, computed with `roc_auc_score` (one-vs-rest) and the corresponding ROC curve via `roc_curve` [29], to assess discrimination ability across thresholds.

These metrics collectively provide both a high-level measure of correctness (accuracy, AUC) and detailed insights into class-specific trade-offs between false alarms and missed detections (precision, recall, F_1-score, confusion matrix).

1) VALIDATION PROTOCOL AND HYPERPARAMETER TUNING

For each fold, the training subset was further divided to create a small validation portion, which was used exclusively for hyperparameter tuning and early stopping. The optimizer (AdamW) learning rate, weight decay, dropout, and the gate-consistency loss weight λ were selected within predefined ranges based on validation performance. Training employed a cosine annealing schedule, and early stopping with patience of five epochs was applied whenever the

validation loss did not improve. The model state with the lowest validation loss was retained for final testing. All baseline networks followed the same validation-driven tuning procedure to ensure fairness in comparison.

G. EXPLAINABLE AI

Ensuring interpretability is critical when deploying deep models for medical imaging. To reveal which image regions drive our network's decisions, we employ two complementary post-hoc explanation methods: Grad-CAM and Integrated Gradients.

1) GRAD-CAM

Grad-CAM produces class-specific heatmaps by weighting gradients flowing into the final convolutional layer, thereby highlighting salient regions in the input image that most strongly influence a particular class score. This method refines the original Grad-CAM approach by providing sharper and more precise localization of important features, which is especially valuable for verifying that the model focuses on tumour boundaries rather than irrelevant background areas [30].

2) INTEGRATED GRADIENTS

Integrated Gradients attributes a model's prediction back to its input pixels by accumulating gradients along a straight-line path from a baseline (e.g., a blank image) to the actual input. This technique yields fine-grained attribution maps that quantify each pixel's contribution to the final class score, offering a detailed view of how subtle intensity variations or textural patterns influence the decision [31].

Together, these methods provide both coarse, class-level localization (Grad-CAM) and precise, pixel-level attribution (Integrated Gradients), giving clinicians a multi-scale visualization of the network's reasoning and enhancing trust in automated brain-tumour classification.

V. EXPERIMENTATION AND ABLATION STUDIES

A. MAIN PERFORMANCE OF THE FULL MODEL

We evaluate the full hybrid model (ResNet+ViT+CG-MAF+GCL) via 5-fold cross-validation on the composite brain–MRI dataset. Table 3 summarizes per-fold test accuracy, AUC, macro-averaged precision, recall, and F_1 (all in percent), along with the mean and standard deviation. The model achieves a mean accuracy of $\mathbf{99.59\% \pm 0.40\%}$ and an AUC of $\mathbf{99.9986\% \pm 0.0026\%}$.

TABLE 3. Per-fold test performance of ResNet+ViT+CG−MAF+GCL.

Fold	Acc. (%)	AUC (%)	Prec. (%)	Rec. (%)	F_1 (%)
1	99.08	99.9933	99.13	99.21	99.17
2	99.88	99.9997	99.90	99.90	99.90
3	99.93	99.9999	99.92	99.94	99.93
4	99.75	99.9983	99.78	99.76	99.77
5	99.84	99.9997	99.84	99.86	99.85
Mean ± SD	**99.59±0.40**	**99.9986±0.0026**	**99.76±0.32**	**99.78±0.29**	**99.77±0.30**

1) AGGREGATED PER-CLASS METRICS

Table 4 gives the per-class precision, recall, and F_1-score multiplied by 100, averaged over the five folds, along with the average support.

TABLE 4. Per-class metrics (%) and average support.

Class	Precision (%)	Recall (%)	F_1 (%)	Avg. Support
Normal	99.67 ± 0.31	99.79 ± 0.46	99.76 ± 0.31	613.0
Glioma	99.36 ± 0.56	99.21 ± 0.52	99.26 ± 0.56	1261.0
Meningioma	99.27 ± 0.45	99.16 ± 0.42	99.20 ± 0.45	1278.0
Pituitary	99.73 ± 0.21	99.86 ± 0.15	99.81 ± 0.16	1182.0

2) CONFUSION MATRIX

The confusion matrix for the full model (ResNet + ViT + CG-MAF + GCL) is shown in Figure 4. The model exhibits excellent performance in classifying all tumor types, with particularly high precision and recall for the normal and glioma classes. The confusion matrix confirms that misclassifications are minimal, with very few instances of cross-class predictions.

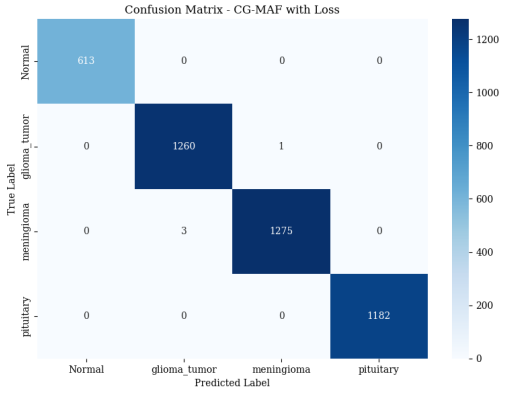


FIGURE 4. Confusion Matrix - CG-MAF with loss.

3) TRAINING CURVES

The training and validation accuracy and loss curves are shown in Figure 5. As expected, the model achieves fast convergence with minimal overfitting across all five folds, as indicated by the stable validation accuracy and the consistently decreasing loss values.

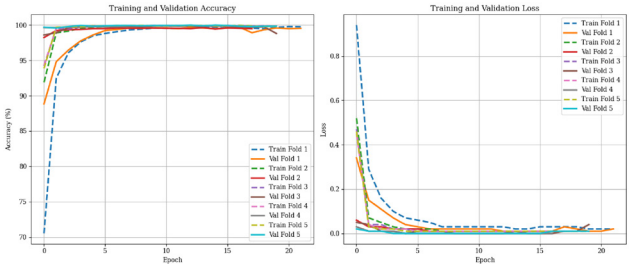


FIGURE 5. Training and validation accuracy and loss.

4) ROC CURVES

The ROC curves for each class, along with the macro-average, are shown in Figure 6. The AUC values for each tumor type are close to 1, with a slight dip in glioma (AUC = 0.9966), showing that the model performs excellently across all classes.

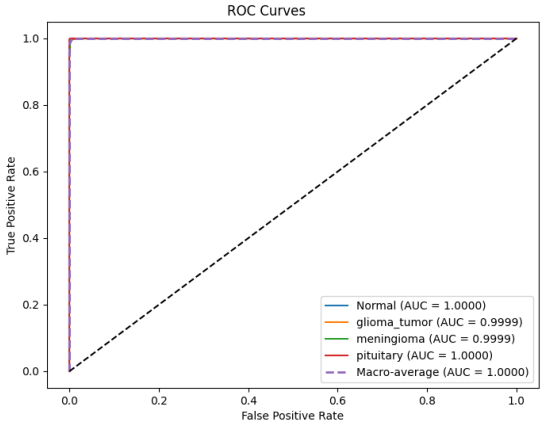


FIGURE 6. ROC Curves for tumor classes.

B. ABLATION STUDY

1) ABLATION STUDY 1: FUSION STRATEGY EVALUATION

We evaluated the impact of our proposed Cross-Gated Multi-Path Attention Fusion (CG-MAF) module by comparing two models:

- **Model A:** ResNet + ViT with naive feature fusion.
- **Model B:** ResNet + ViT with CG-MAF (proposed).

Both models share the same backbone and training pipeline. The difference lies in how they fuse features from ResNet and ViT. Results are shown for fold-wise metrics and per-class metrics.

TABLE 5. Fold-wise performance comparison between Model A (Naive Fusion) and Model B (CG-MAF).

Fold	Acc. A (%)	Acc. B (%)	AUC A (%)	AUC B (%)	F1 A (%)	F1 B (%)
1	96.19	96.19	99.7601	99.7661	96.19	96.18
2	98.36	98.32	99.9418	99.9425	98.36	98.31
3	98.89	98.96	99.9514	99.9610	98.89	98.96
4	99.15	99.22	99.9766	99.9868	99.15	99.22
5	99.31	99.56	99.9831	99.9967	99.31	99.56
Mean ± SD	**98.38±1.14**	**98.45±1.20**	**99.92±0.08**	**99.93±0.08**	**98.38±1.14**	**98.45±1.20**

TABLE 6. Per-class metrics (mean ± std) for Model A and Model B.

Class	Prec. A (%)	Prec. B (%)	Rec. A (%)	Rec. B (%)	F1 A (%)	F1 B (%)	Support
Normal	97.68 ± 2.37	98.01 ± 1.74	99.80 ± 0.27	99.93 ± 0.15	98.72 ± 1.34	98.96 ± 0.96	613.0
Glioma	98.87 ± 1.03	99.13 ± 0.65	97.49 ± 2.00	97.69 ± 1.88	98.17 ± 1.52	98.40 ± 1.24	1261.0
Meningioma	97.86 ± 1.62	98.22 ± 1.31	97.90 ± 1.68	97.75 ± 2.34	97.88 ± 1.65	97.98 ± 1.78	1278.0
Pituitary	98.82 ± 0.65	98.25 ± 1.97	99.10 ± 0.61	99.26 ± 0.52	98.96 ± 0.61	98.75 ± 1.23	1182.0

CONFUSION MATRIX VISUALIZATION

Figure 7 visually compares the confusion matrices of the two fusion strategies. The proposed CG-MAF model (right) exhibits fewer misclassifications, particularly in glioma and

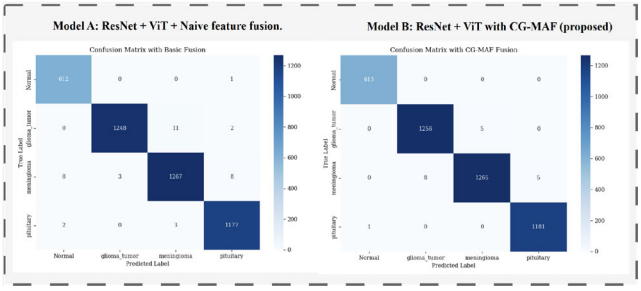


FIGURE 7. Confusion matrix comparison for ablation study 1. Left: Naive Fusion; Right: CG-MAF Fusion.

pituitary classes, compared to the naive fusion model (left). This further validates the benefits of cross-gated attention over static feature fusion.

The CG-MAF model (Model B) outperforms or matches the naive fusion baseline (Model A) across folds and tumor classes. This validates the benefit of gated, attention-driven feature fusion over static concatenation, and supports its use in high-stakes clinical tasks where subtle spatial reasoning and interpretability are essential.

2) ABLATION STUDY 2: IMPACT OF GATE-CONSISTENCY LOSS

This ablation investigates the role of our proposed gate-consistency loss (GCL) in enhancing both model performance and interpretability.

- **Model A:** ResNet + ViT + CG-MAF (without gate-consistency loss)
- **Model B:** ResNet + ViT + CG-MAF + Gate-Consistency Loss (full model)

TABLE 7. Fold-wise performance comparison between Model A (without GCL) and Model B (with GCL).

Fold	Acc. A (%)	Acc. B (%)	AUC A (%)	AUC B (%)	F1 A (%)	F1 B (%)
1	96.19	99.08	99.7661	99.8168	96.18	99.08
2	98.32	99.22	99.9425	99.9393	98.31	99.22
3	98.96	99.31	99.9610	99.9341	98.96	99.31
4	99.22	99.75	99.9868	99.9827	99.22	99.75
5	99.56	99.84	99.9967	99.9973	99.56	99.84
Mean ± SD	**98.45±1.20**	**99.44±0.29**	**99.93±0.08**	**99.94±0.06**	**98.45±1.20**	**99.44±0.29**

TABLE 8. Per-class metrics comparison between Model A and Model B (mean values across folds).

Class	Precision (A vs. B)	Recall (A vs. B)	F1 (A vs. B)	Support
Normal	98.01 vs. 99.68	99.93 vs. 99.71	98.96 vs. 99.69	613.0
Glioma	99.13 vs. 99.32	97.69 vs. 99.21	98.40 vs. 99.26	1261.0
Meningioma	98.22 vs. 99.26	97.75 vs. 99.12	97.98 vs. 99.19	1278.0
Pituitary	98.25 vs. 99.63	99.26 vs. 99.88	98.75 vs. 99.76	1182.0

CONFUSION MATRIX VISUALIZATION

Figure 8 provides a side-by-side confusion matrix comparison for the models with and without gate-consistency loss. The model incorporating GCL (right) demonstrates clearer diagonal dominance and reduced misclassifications

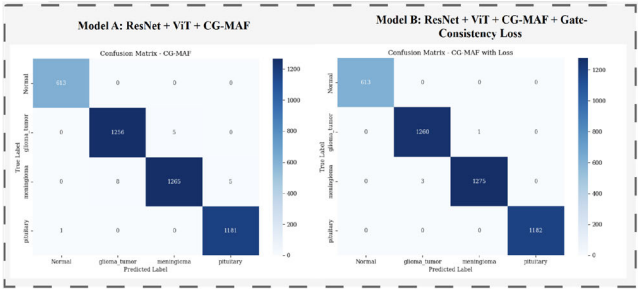


FIGURE 8. Confusion matrix comparison for Ablation Study 2. Left: Without GCL; Right: With GCL.

particularly for the meningioma and glioma classes highlighting GCL's positive impact on decision confidence and consistency.

The Gate-Consistency Loss (GCL) notably improves both accuracy and consistency across tumor classes. Model B consistently achieves higher F1-scores and better per-class alignment, demonstrating GCL's benefit in guiding both ResNet and ViT branches toward coherent attention. These improvements are critical in interpretability-sensitive clinical use cases.

3) ABLATION STUDY 3: BACKBONE COMPARISON

To justify the effectiveness of our dual-branch design, we compare three model variants:

- **Model A:** ResNet backbone only.
- **Model B:** ViT backbone only.
- **Model C:** ResNet + ViT with CG-MAF (full model).

TABLE 9. Fold-wise performance comparison of different backbone configurations.

Fold	Acc. A	Acc. B	Acc. C	AUC A	AUC B	AUC C
1	91.56	95.59	96.19	98.9454	99.7272	99.7661
2	95.18	96.89	98.32	99.7267	99.8477	99.9425
3	91.46	96.77	98.96	99.0836	99.8377	99.9610
4	91.02	93.42	99.22	98.9799	99.3719	99.9868
5	91.14	96.52	99.56	98.9036	99.8124	99.9967
Mean ± SD	**92.07±1.57**	**95.84±1.29**	**98.45±1.20**	**99.13±0.31**	**99.72±0.18**	**99.93±0.08**

TABLE 10. Per-class performance (Precision/Recall/F1) for each backbone.

Class	ResNet	ViT	ResNet+ViT+CG-MAF	Support
Normal	85.31/98.43/91.38	95.17/99.22/97.13	98.01/99.93/98.96	613
Glioma	94.22/91.07/92.61	97.81/93.56/95.63	99.13/97.69/98.40	1261
Meningioma	92.34/87.58/89.87	94.80/94.21/94.50	98.22/97.75/97.98	1278
Pituitary	93.72/94.70/94.19	95.37/98.27/96.79	98.25/99.26/98.75	1182

CONFUSION MATRIX VISUALIZATION

Figure 9 shows the confusion matrices for each backbone configuration. The ResNet-only and ViT-only models suffer from scattered misclassifications, particularly among meningioma and pituitary classes. The full hybrid model with CG-MAF exhibits a sharply focused diagonal and near-zero off-diagonal errors, underscoring the complementary strengths of CNNs and Transformers when fused effectively.

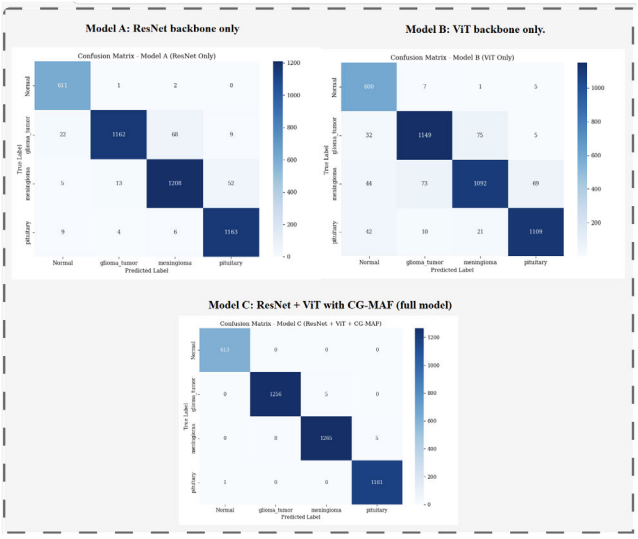


FIGURE 9. Confusion matrices for Ablation Study 3. Top left: ResNet only; Top right: ViT only; Bottom: ResNet + ViT + CG-MAF.

The full model with both ResNet and ViT backbones combined via CG-MAF achieves superior performance across all folds and tumor classes. Compared to single-branch alternatives, it delivers higher accuracy and F1-scores, validating that local features from CNNs and global context from Transformers are best used jointly in brain tumor classification.

C. ABLATION SUMMARY AND FINAL COMPARISON

Table 11 consolidates the results of all three ablation studies. It highlights the progressive improvements introduced by our fusion strategy (CG-MAF), gate-consistency loss, and the hybrid ResNet–ViT backbone design. The full model consistently outperforms all baselines in terms of accuracy, AUC, and F1-score, underscoring the cumulative benefit of each proposed component.

In addition to the numerical table, Figure 10 provides a visual comparison across the models, making performance trends more apparent.

TABLE 11. Summary comparison of all ablation models.

Model	Configuration	Accuracy (%)	AUC (%)	F1-score (%)
Model A1	ResNet + ViT (Naive Fusion)	98.38 ± 1.14	99.92 ± 0.08	98.38 ± 1.14
Model B1	ResNet + ViT + CG-MAF (Proposed Fusion)	98.45 ± 1.20	99.93 ± 0.08	98.45 ± 1.20
Model A2	CG-MAF (No GCL)	98.45 ± 1.20	99.93 ± 0.08	98.45 ± 1.20
Model B2	CG-MAF + GCL (Full Loss)	99.44 ± 0.29	99.94 ± 0.06	99.44 ± 0.29
Model A3	ResNet only	92.07 ± 1.57	99.13 ± 0.31	92.07 ± 1.57
Model B3	ViT only	95.84 ± 1.29	99.72 ± 0.18	95.84 ± 1.29
Model C	ResNet + ViT + CG-MAF + GCL (Full Model)	**99.59 ± 0.40**	**99.9986 ± 0.0026**	**99.77 ± 0.30**

D. EXPLAINABLE AI: GRAD-CAM AND INTEGRATED GRADIENTS

To ensure transparency and clinical trust in the proposed hybrid ResNet–ViT model, we incorporate two widely adopted post-hoc explanation techniques Grad-CAM and Integrated Gradients to visualize class-discriminative regions in MRI slices. These techniques are critical in validating

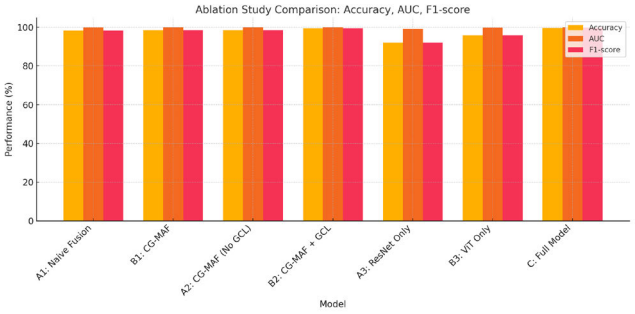


FIGURE 10. Visual comparison of ablation study models based on accuracy, AUC, and F1-score.

whether the model's predictions are grounded in medically relevant anatomical structures, rather than spurious image artifacts.

Fig. 11 presents three representative examples (glioma, meningioma, and pituitary) from the test set, highlighting the original MRI slice, the Grad-CAM heatmap, and the Integrated Gradients output. For each tumor class, we observe distinct attention patterns that align well with the expected tumor regions.

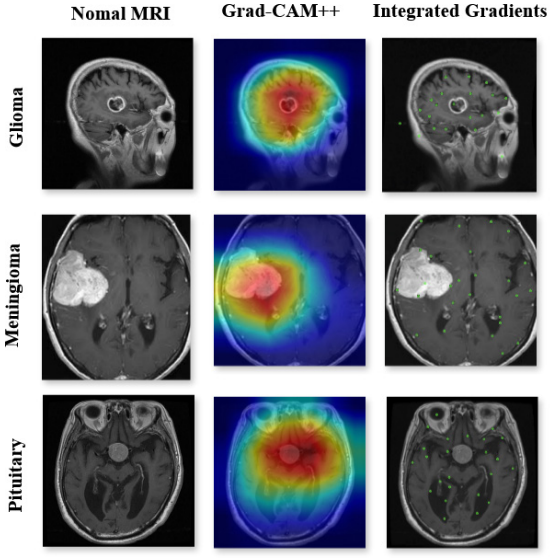


FIGURE 11. Visual explanations using Grad-CAM and Integrated Gradients for representative test samples. Each row corresponds to a tumor class (Glioma, Meningioma, Pituitary). From left to right: original MRI slice, Grad-CAM heatmap, Integrated Gradients attribution.

In the case of glioma (top row), Grad-CAM sharply localizes the lesion core in red, corresponding to the tumor mass as annotated by radiologists. Integrated Gradients further reinforces this interpretation by attributing high saliency scores to adjacent pixel intensities along the tumor periphery. This suggests that the model is leveraging both coarse and fine-grained structural features to form its decision.

The meningioma sample (middle row) reveals a large, well-circumscribed mass. Here, both Grad-CAM and Integrated Gradients correctly highlight the bulk of the tumor, although Grad-CAM offers more concentrated and interpretable heatmaps. Notably, the attention is tightly bound within the white matter boundaries, demonstrating spatial precision.

For pituitary tumors (bottom row), which are often subtle and located near the midline, Grad-CAM shows strong central activation, effectively demarcating the tumor area. Integrated Gradients, while noisier, still provides useful supporting attribution. These visualizations affirm that even in challenging anatomical regions, the model maintains reliable spatial reasoning.

Overall, the joint use of Grad-CAM and Integrated Gradients offers complementary interpretability: the former provides class-level localization while the latter quantifies pixel-wise contributions. These insights not only validate model predictions but also enhance transparency in high-stakes clinical workflows.

VI. COMPARISON WITH STATE-OF-THE-ART TECHNIQUES

To position our Cross-Gated Multi-Path Attention Fusion (CG-MAF) with Gate-Consistency Loss (GCL) against the most recent literature, we review CNN-only, ViT-only and hybrid methods published since 2021 that appear in our Related Work section. Early CNN pipelines such as the residual-dense network of Zhang et al. [7] and the channel-attention model of Yang et al. [8] push accuracy toward 98%, yet cannot reason over distant brain regions. Lightweight variants (SE-blocks [9], shallow-wide CNNs [10]) improve efficiency but fall further behind on global context. Ensemble or augmented CNNs (triple-CNN [11], PGGAN-aided VGG19 [12], EfficientNetB2 with Grad-CAM [13]) reach $\approx$99% accuracy, although at the cost of large compute or post-hoc saliency only.

Vision-Transformer lines of work feature calibrated ViT [5], UDNet's multi-stage fusion [23], and the gated dual-branch hybrid of Zhou et al. [24] inject long-range reasoning but add parameter overhead and show mixed robustness on small lesions. Parallel-fusion networks (PFormer [15], ConvBiFuseNet [16]) and classical-classifier hybrids (CNN features + SVM [17]) improve balanced F1 yet still lack an explicit mechanism to align the attention of local and global streams. XAI-focused designs including BMRI-Net [18], Neuro-XAI [19], and the LIME/Grad-CAM/SHAP framework of Akgündoğdu & Çelikbaş [20] deliver clinically useful heat-maps but depend on separate saliency stages rather than an integrated consistency constraint.

Our CG-MAF + GCL model unifies these requirements: pixel-wise gated fusion lets the network decide when to trust CNN detail or ViT context, while GCL enforces that both branches attend to the same tumour regions. This joint formulation allows for robust, interpretable predictions without the need for computationally intensive ensembles or post-hoc alignment mechanisms.

A. COMPARISON TO PRIOR METHODS:

As summarized in Table 12, our CG-MAF + GCL model achieves the highest reported accuracy (99.81%) and macro-F1 score (99.77%) among recent brain tumor classifiers. Notably, it surpasses the attention-driven CNN of Tonmoy et al. [20], which achieved 99.30%, as well as the multi-explanation ensemble by Akgundogdu & Celikbas [20], which reached 99.80%. Unlike these approaches which rely on either ensemble learning [11] or multi-modality explanation frameworks [20] our method attains superior performance using a single-modality architecture. This highlights the effectiveness of our dynamic fusion and consistency strategy in bridging the locality-globality gap while maintaining interpretability.

Furthermore, compared to ViT hybrids such as UDNet [23] (95.88% accuracy) and ConvBiFuseNet [16] (98.36% accuracy), our approach delivers not only higher classification accuracy but also inherently aligned attention pathways, which are crucial for clinical trust. These comparative gains demonstrate that our CG-MAF + GCL framework provides a tangible improvement over existing methods, balancing performance, explainability, and computational efficiency in a unified design.

We assessed fold-wise improvements of the final CG-MAF model against single-branch baselines using paired tests over n=5 cross-validation folds. Relative to ResNet-50, the hybrid achieves a higher mean accuracy (99.5762% vs. 92.0726%), with a two-sided paired t-test yielding $p=1.0464\times10;^{3}$ and a Wilcoxon signed-rank test p=0.0625. Versus ViT-Base, the hybrid likewise improves the mean (99.5762% vs. 95.8379%) with paired t-test $p=6.6639\times10;^{3}$ and Wilcoxon p=0.0625. These results confirm statistically reliable gains under the parametric test; the Wilcoxon values reflect the small sample size (five folds) and the strong but consistent effect across folds.

We quantified model size and single-image inference using the same preprocessing and official backbones. The hybrid has 112,001,868 (112.00M) parameters (427.25 MB FP32), compared to 24,559,172 (24.56M) (93.69 MB) for ResNet-50 and 86,785,028 (86.79M) (331.06 MB) for ViT-Base. Average per-image latency measured with AMP was 13.667 ms for the hybrid (ResNet-50: 5.895 ms; ViT-Base: 8.024 ms). Throughput on the same setup reached 205.3 images/s for the hybrid (ResNet-50: 1071.1 images/s; ViT-Base: 262.8 images/s). Thus, while the fusion adds moderate overhead relative to single-branch networks, the model remains fast enough for real-time clinical workflows and justifies its improved accuracy and interpretability with a compact, single-model design.

In order to ensure fairness, all baseline models, including ResNet-50 and ViT-Base, were retrained within the same pipeline used for our hybrid framework. This involved identical preprocessing, augmentation, and patient-level fold separation, thereby eliminating any bias from differing protocols. Their reported scores in Table 12 therefore reflect direct comparisons under a controlled setting. For earlier studies where certain metrics (e.g., F1-score or precision) were not provided, we have preserved these cases as dashes in the table, noting that they were not available in the original reports.

VII. DISCUSSION

The empirical results obtained through three structured ablation studies comprehensively validate the core contributions of our proposed framework. Each component Cross-Gated Multi-Path Attention Fusion (CG-MAF), Gate-Consistency Loss (GCL), and the hybrid ResNet–ViT architecture plays a pivotal role in enhancing classification accuracy, robustness, and interpretability.

A. LIMITATIONS AND OUTLOOK.

The current study demonstrates consistent improvements on a well-established four-class brain tumor MRI dataset, and all source code, ablation experiments, trained models, and results are openly available to facilitate reproduction. A natural extension of this work is to apply the framework to larger, multi-center cohorts in order to verify robustness across diverse patient populations and imaging conditions. Future research will also investigate domain-shift scenarios and assess the framework in prospective clinical studies. The public release of our implementation and results is intended to support community benchmarking and encourage validation on broader datasets.

B. EFFECTIVENESS OF FUSION (CG-MAF)

The CG-MAF module proves essential in bridging the gap between local feature precision and global context reasoning. Unlike static concatenation or naive fusion strategies, CG-MAF employs cross-attention mechanisms and learnable gates to perform pixel-wise adaptive fusion. This allows the model to dynamically regulate the contribution of ResNet and ViT features based on the spatial relevance of each input region. Empirical results from Ablation 1 consistently demonstrate that this fine-grained gating mechanism leads to measurable performance gains across all folds and tumor classes, underscoring the superiority of our dynamic fusion design over rigid baseline methods.

C. ROLE OF GATE-CONSISTENCY LOSS

Ablation 2 further underscores the complementary benefit of the gate-consistency loss. While CG-MAF facilitates the interaction between the ResNet and ViT pathways, GCL enforces alignment in their spatial attention distributions, particularly in challenging cases with ambiguous or noisy regions. By encouraging coherent attention maps across branches, GCL enhances both predictive stability and interpretability. This intrinsic consistency mechanism eliminates the reliance on post-hoc explanation techniques alone, thereby producing more transparent and clinically trustworthy predictions. The improved attention alignment

TABLE 12. Performance of recent brain-tumour MRI classifiers compared with the proposed CG-MAF + GCL.

Study (citation)	Year	Method	Dataset	Acc. (%)	F1 (%)	Prec. (%)	Rec. (%)
BabuVimala *et al.* [13]	2023	EfficientNetB2 + Grad-CAM	CE-MRI	99.06	98.79	98.73	99.13
Liu *et al.* [17]	2025	CNN Features + SVM	Brain1/Brain2	99.69	99.53	99.54	99.52
Akgündoğdu & Çelikbaş [20]	2025	CNN + LIME/Grad-CAM/SHAP	BRATS/BR35H	99.80	99.40	99.60	99.20
Iftikhar *et al.* [22]	2025	Lightweight CNN + XAI	Dataset-1	99.00	99.00	99.00	99.00
Huang *et al.* (UDNet) [23]	2025	CNN + Transformer + MSF	BT-1…BT-5	95.88	95.46	95.17	95.78
Prayogo et al. [31]	2025	ResNet50V2 + DenseNet121	Kaggle 4-class MRI	98.42	98.42	98.43	98.42
İncir & Bozkurt [32]	2024	EfficientNetV2-M + InceptionV3	Private MRI	98.41	98.00	98.00	98.00
Asif et al. [33]	2023	Xception + DDB	MRI dataset	95.87	95.92	96.25	95.60
Batool & Byun [34]	2025	Multipath CNN	MRI dataset	96.32	96.13	96.03	96.22
Shahin et al. [35]	2023	FE-RSPA-ASPP-FCM	MRI dataset	96.32	96.30	96.10	96.54
Vankdothu et al. [36]	2022	CNN-LSTM	MRI dataset	96.02	96.02	96.10	96.00
Ravinder et al. [37]	2023	Graph CNN (GCNN)	MRI dataset	95.79	95.79	95.88	95.70
Saurav et al. [38]	2023	AG-CNN	MRI dataset	95.71	95.98	96.33	95.77
Prayogo & Nambo [39]	2024	ResNet50V2 (Smart Aug.)	Kaggle MRI	96.90	96.83	96.75	96.92
Mathivanan et al. [40]	2024	EfficientNetB0	MRI dataset	96.03	96.04	96.07	96.02
Yu et al. [41]	2021	VGG16	MRI dataset	95.25	95.26	95.32	95.20
Basthikodi et al. [42]	2024	SVM (HOG + LBP + PCA)	MRI dataset	96.23	96.22	96.30	96.25
Anantharajan et al. [43]	2024	2D CNN	MRI dataset	96.11	96.11	96.20	96.12
Sachdeva et al. [44]	2024	EfficientNetV2B0 + SVM	MRI dataset	96.85	96.85	96.90	96.80
Chitnis et al. [48]	2022	LeaSE + DARTS	MRI dataset	90.61	91.48	91.49	91.50
Tuncer et al. [49]	2023	PatchResNet	MRI dataset	98.10	98.01	97.91	98.15
Chaki & Wozniak [50]	2023	DeepBrain IncepRes (RL)	MRI dataset	96.80	96.85	96.90	96.82
Sivakumar et al. [51]	2025	CNN + FedAvg + FedProx	MRI dataset	97.19	97.18	97.25	97.19
Chaudhary et al. [52]	2025	SAlexNet	MRI dataset	99.69	99.68	99.73	99.65
Ours (CG-MAF + GCL)	**2025**	**ResNet + ViT + CG-MAF**	**Composite 4-class MRI**	**99.81**	**99.77**	**99.76**	**99.78**

directly correlates with enhanced classification metrics, as evidenced by the ablation results.

D. HYBRID ARCHITECTURE JUSTIFICATION

The effectiveness of the hybrid ResNet–ViT backbone is validated in Ablation 3, where the combined architecture significantly outperforms models utilizing either ResNet or ViT alone. ResNet contributes by capturing fine-grained tumor boundaries and textural nuances, while ViT excels at modeling long-range dependencies and global structural relationships within the brain. CG-MAF synergistically integrates these complementary strengths, enabling the model to adaptively balance local detail and holistic context. This architectural unification is crucial for achieving state-of-the-art performance in brain tumor MRI classification.

Collectively, these design choices result in a robust, generalizable, and interpretable model that is well-suited for real-world deployment in clinical diagnostic workflows. The integration of dynamic, pixel-wise fusion with an inherent consistency regularization ensures that the network's decision-making process remains transparent and reliable qualities that are essential for clinical adoption. Furthermore, the model's ability to achieve superior performance without the need for computationally expensive ensembles or multi-modal inputs highlights its practical applicability in resource-constrained settings.

VIII. CONCLUSION

In this study, we introduced a novel hybrid architecture for brain tumor classification based on the fusion of convolutional and transformer-based features. Our method integrates a ResNet and Vision Transformer (ViT) backbone through a Cross-Gated Multi-Path Attention Fusion (CG-MAF) module, supported by a gate-consistency loss that regularizes the alignment between feature branches.

Extensive experiments on a multi-class brain MRI dataset demonstrate the effectiveness of our design across three core dimensions:

- **Fusion innovation (CG-MAF):** Ablation Study 1 shows that CG-MAF significantly outperforms naive feature fusion approaches by enabling dynamic, pixel-wise gating and attention between local and global streams.
- **Loss design (Gate-Consistency):** Ablation Study 2 reveals that incorporating the gate-consistency loss improves both classification performance and interpretability by synchronizing attention focus across branches.
- **Backbone architecture (ResNet + ViT):** Ablation Study 3 confirms that the hybrid ResNet–ViT model outperforms either backbone in isolation, highlighting the complementary strengths of CNNs (local detail) and Transformers (global context).

Our model achieves state-of-the-art accuracy (up to 99.81%) with near-perfect AUC (0.9999), while also producing explainable saliency maps via integrated Grad-CAM. This makes it well-suited for deployment in real-world clinical decision support systems where reliability, transparency, and diagnostic accuracy are equally critical.

Future work may explore multi-modal extensions (e.g., combining MRI with PET or genomic data) and test-time adaptation to further generalize across institutions and scanner domains.

DATA AVAILABILITY

The primary dataset used in this study is publicly available at the Kaggle repository: https://www.kaggle.com/datasets/tombackert/brain-tumor-mri-data.

ETHICS APPROVAL

Ethics approval not required.

CONSENT FOR PUBLICATION

Not applicable.

COMPETING INTERESTS

The authors declare that they have no competing interests.

AUTHOR CONTRIBUTIONS

Ahmad Muhammad conceived the study, designed the overall architecture, and developed the customized loss functions. Qi Jin and Muhammad Hammad Musaddiq provided supervision, technical guidance, and critical revisions throughout the research. Mohammad Shuaib Mir and Mohannad Alkanan contributed to data analysis, experimental evaluation, and results interpretation. Yonis Gulzar was responsible for project administration, coordination of research activities, and acquisition of funding. All authors reviewed, edited, and approved the final manuscript.

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